

Comparing ChatGPT and Humans on World Knowledge and Common-sense Reasoning Tasks: A case study of the Japanese Winograd Schema Challenge

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ABSTRACT

Large Language Models (LLMs) like ChatGPT have great potential to influence the way people search for information and make decisions. Those effects are already felt in many fields, so it's necessary to get a better understanding of how they work and how their linguistic competencies and decision-making heuristics compare to humans. Benchmark tests on these skills have been primarily developed in English. Due to disparities in training data in different languages, LLM performance in English may not necessarily translate to other languages. It is essential to expand the domain of evaluation beyond English. In this study we evaluate ChatGPT performance on world knowledge and commonsense reasoning tasks vis-à-vis human performance in Japanese. Our results show that ChatGPT achieves lower accuracy compared to humans. We also report differences in naturalness judgements. These results shed light on the differences between humans and LLMs and can inform future studies.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*; **Empirical studies in HCI**;

KEYWORDS

NLU Benchmark tests, Winograd Schema Challenge, Japanese, ChatGPT (GPT-3.5)

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1 INTRODUCTION

Until recently, the ability to use commonsense reasoning and world knowledge was a far-off goal for computer-based systems but came naturally to humans. New generation Large Language Models

(LLMs) increasingly demonstrate what seems to be the capability of human-like reasoning and they have great potential to affect how people search for information and make decisions. They are already contributing to a variety of sectors including education, journalism, science, etc., but before they can be employed further, we need to get a better understanding of how they work and how their linguistic competencies and decision-making heuristics compare to that of humans.

In response to this need, researchers across a range of disciplines are evaluating LLM performance vis-à-vis human performance on tasks that involve knowledge about the world and common-sense reasoning. For example, [10] investigated counterfactual reasoning in LLMs and concluded that they do not achieve human accuracy level. Similarly, [8] evaluated LLMs on planning and explanation generation and found that they underperform compared to humans (see also [25] on planning). More recently, [7] studied LLMs' ability to draw inferences, based on how information is presented in discourse, and found that LLM performance is comparable to human results. Similarly, [11] investigated LLMs' generalized event knowledge, and their ability to reason about possible vs. impossible and likely vs. unlikely events. They found that LLM performance parallels human judgements, but that unlike humans, LLMs are also affected by linguistic properties of the target sentences, such as voice (active vs. passive), word frequencies, and sentence length.

Despite a growing interest in this emergent area of research, the majority of work is evaluating LLMs using the English language. However, in order for LLMs to fulfil their promise and avoid the disparity in distribution and usage of language technologies, testing in other languages is needed [5]. In this work we address this need by using Japanese as the language of evaluation. Establishing a baseline for Japanese is important because there is an imbalance in LLM training data. Specifically, while GPT-3 is pre-trained on 119 languages [6], 93% of this data is in English [1]. Preliminary evidence from MEGA, a model for Multilingual Evaluation of Generative AI [1], which studies LLM performance in 70 typologically different languages, suggests that the amount of data in a particular language available during LLM pre-training might affect performance. For example, the authors found that on Paraphrase Identification tasks, where the system must decide whether two different paraphrases express the same meaning, OpenAI models perform much better in French (72.1% accuracy) than in Japanese (67% accuracy). They attribute the difference in performance to the disparity in the amount of pre-training data: 3.5B tokens in French compared to only 240M tokens in Japanese. These findings suggest that LLM successes in one language might not necessarily translate into another language.

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This paper contributes to the emergent field of LLM evaluation by studying how well ChatGPT performs on the Winograd Schema Challenge, a benchmark test for LLMs that evaluates world knowledge. While the majority of the work is conducted in English, we focus on Japanese. The results of two experimental studies, one with human participants and another with ChatGPT (GPT-3.5), demonstrate that ChatGPT underperforms compared to humans, and that surface linguistic features do not seem to affect accuracy. Our contribution is two-fold. First, our pilot study develops criteria for WSC test material selection for Japanese and establishes a preliminary human baseline for this language, which can be used in subsequent work on LLMs. Second, it contributes to the discussion of the validity of standard benchmark tests originally designed for English and translated into other languages.

2 THE WINOGRAD SCHEMA CHALLENGE

The Winograd Schema Challenge (WSC) is a short reading comprehension task originally presented in 2011 as an alternative to the Turing test to determine if the computer system demonstrated an ability to think comparable to human intelligence. A Winograd Schema consists of one sentence and one question as in the following:

(1) The trophy would not fit into (2) the brown suitcase
because (3) it was too big. What was too big?

The sentence introduces two entities (1, 2) and then uses a pronoun (3) to refer back to one of the entities. The question requires the system to identify the referent of the pronoun, a task known as pronoun resolution. Correctly answering the question requires use of commonsense reasoning and world knowledge [13].

Winograd Schemas must be developed carefully to avoid being too simple that they could be answered by pragmatic or syntactic clues alone and they should not be too difficult that a native English-speaking adult might misunderstand. Winograd Schemas should also be *google-proof* and avoid questions that can be answered by word co-occurrence patterns. In addition to following these restrictions and basic structure, the sentence-question pair must also have a single word that can be changed to flip the correct answer. This is to ensure that the answer to the question is not determinable by pragmatics or frequency statistics. Table 1 shows a pair of WSC sentences.

Many notable Winograd Schema collections have been developed with the goal of testing NLP systems on pronoun resolution such as Winogrande [19], The Definite Pronoun Resolution (DPR) set [18], The Pronoun Disambiguation Problem Dataset [15], and The Winograd Natural Language Inference set, which appears in the GLUE [27] and SuperGLUE [26] benchmark tests. Many of them were created with relaxed requirements due to the difficulty of crafting items that fit all the original stipulations. Modifications to the original requirements are also often introduced when the WSC is translated into other languages. The Shibata test set in Japanese [21] is a translation of the DPR set, which follows the WSC conventions but with relaxed restrictions on the *google-proof* rule, allowing for items that might be answerable by correlation statistics or pragmatic reasoning. An example test sentence from the Shibata set is shown below. One major difference from English is that in Japanese, the information is packaged into two independent

sentences. The first sentence introduces the two event participants (the professor and the student), and the second sentence contains the pronoun that needs to be resolved.

教授は学生に低い成績をつけた。彼は厳格だったからだ。

The professor gave the student a bad grade because he was strict.

The Shibata set became the standard test bed for Japanese and was used in the Japanese Feature Change Knowledge Base (JFCKB) [16], the development of XWINO [24] and a comparison of pre-trained language models [23]. This is the version of the WSC challenge that we use here.¹

Despite the popularity of the WSC as a tool for evaluating Natural Language technology, little is known about human performance. The creators of WSC sets often stipulate that questions must be easily answerable by a human, but many do not establish a gold standard using human data, nor did studies using benchmark tests such as GLUE and SuperGLUE emphasize comparison with human performance. While some efforts have been made to establish human baselines [17], most of the research uses English benchmark tests. The new generation of LLMs necessitates the establishment of human baselines in many languages, due to human-like performance on many cognitive and linguistic tasks (cf. [11], [25], [8], [10]).

In what follows, we report results of two studies. In Study 1, we collect human judgements on the WSC in Japanese, thus establishing a preliminary baseline. Study 2 evaluates the performance of ChatGPT on the same tasks.

3 EXPERIMENTAL STUDIES

We conduct two studies. The first is with human participants and the second is with ChatGPT. Both human participants and ChatGPT see the same stimuli from the Japanese WSC and complete the same tasks. Task 1 asks the subject to resolve the WSC problem. Task 2 asks the subject to rate the stimuli for naturalness in Japanese.

3.1 Study 1: Human performance on the Japanese Winograd Schema Challenge

The goal of this study was to establish a human baseline for the Japanese WSC. While the original WSC assumed 100% accuracy by humans, evaluation of the English test sentences by [3] later revealed that humans achieve 92% accuracy on the test. No comparable study exists for Japanese.

When evaluating the WSC in Japanese, an important additional factor that might affect the result is the question of naturalness. Due to significant typological differences between English and Japanese, translation of Winograd Schema test sentences into Japanese presents a challenge. Researchers who translated the WSC into Japanese acknowledge that faithfulness to the original syntax and the test goals often yields unnatural results in Japanese [21]. However, the test was widely applied regardless of these typological challenges. When evaluating human data, we want to evaluate not only the accuracy but also investigate how natural these examples

¹Another version of the WSC test in Japanese was developed by Tanaka et al.[9], but to our knowledge, studies on Japanese WSC have used the Shibata set and no published work has used the Tanaka translation.

Table 1: Example of Winograd Schema Challenge

Question	Answer	Special Word
The trophy would not fit into the brown suitcase because it was too big. What was too big?	the trophy	big
The trophy would not fit into the brown suitcase because it was too small. What was too small?	the brown suitcase	small

sound to native speakers and whether naturalness might affect performance. Therefore, in the study, participants performed two tasks:

- (1) pronoun resolution task (the essence of the WSC that tests world knowledge and reasoning), but also
- (2) naturalness judgement task (rating how natural the examples were).

3.1.1 Participants. 33 native speakers of Japanese participated in the study. The average age was 54.7. There were 10 men, 20 women and 3 participants who selected “other” or declined to state. Participants were recruited via academic snowball sampling with three starting nodes.

3.1.2 Design. The studies had two conditions. Stimuli in condition 1 were unmodified Japanese WSC problems (Overt pronoun condition). In condition 2, the pronoun was omitted (Null pronoun condition). This condition was created because Japanese is considered a pro-drop language [22]. In pro-drop languages, a sentence may omit a pronoun which still has a real-world referent. The omitted pronoun is often referred to as a null pronoun. See Table 2 for example sentences in both conditions. Previous findings about LLM sensitivity to surface sentence features necessitates closer attention to possible structural variations, and the presence vs. absence of an overt pronoun addresses this consideration [11].

3.1.3 Materials. The problems used in the study come from the Shibata translation [21] of the WSC into Japanese. The test consists of 941 question pairs, from which we selected 10 to be used in our study materials. Our selection process was guided by structural and semantic considerations and those of cultural appropriateness. A detailed pre-study analysis of the dataset revealed that many of the items are direct translations from English and that interpretation of these sentences requires specific cultural knowledge (e.g. American football terminology) which may make them impossible to answer for people unfamiliar with the topic, thus violating a foundational criterion of the WSC. Moreover, there were structural inconsistencies (some sentences were structurally complex, while others had only one clause), some had multiple pronouns, while others had mismatch in voice (active vs. passive). These complexity factors should not affect a participant’s ability to answer the question, but they add extra variables to the design which would be difficult to analyze thoroughly with our small dataset. To address these issues, we developed a set of criteria that needed to be met for the problem to be included in our list of stimuli.

- (1) The problem must consist of two simple sentences with no subordinate clauses.
- (2) The problem must not contain any long foreign names in Katakana (Japanese alphabet used to write foreign words)

or culturally specific knowledge that is not familiar to most Japanese people.

- (3) The target referent of the pronoun must either be the subject, object, or indirect object of the first sentence.
- (4) There must be only one pronoun targeted for resolution.
- (5) The voice must be consistent between the first sentence and the second sentence (active-active, or passive-passive).

We generated a random sample from the data and the first 10 question pairs in which both problems fulfilled all five requirements were selected. See appendix A.1 for examples of items which were rejected.

Next, a question to serve as a prompt for pronoun resolution was created for each problem. Like the English WSC, the same pattern was used for all questions for consistency. Finally, a null pronoun version was created for each problem. This was done by simply omitting the pronoun and related particle from the second sentence of the problem. As an example, Table 2 shows both the null and pronoun versions of a question pair used in the experiments. See Appendix A.2 for a full list of our stimuli.

3.1.4 Procedure. The survey was distributed via anonymous link using the online survey platform Qualtrics. Participants were first asked to provide implied consent and verify that they were over 18 years of age and native Japanese speakers. Next, they completed two tasks. Each task had 40 questions (10 pairs, 4 versions each), which were presented in random order. Each participant saw all 40 questions. Task 1 tested accuracy in resolving the ambiguous pronoun. Task 2 asked for a rating of naturalness of the same stimuli sentences from task 1. No filler items were used and in order to mirror the zero-shot approach used with ChatGPT, human participants were not given any training session. See Appendix 3 for instructions given for each task.

In task 1, participants were shown one WSC problem at a time and asked to select the most appropriate answer. Each problem consisted of two Japanese sentences, one prompting question and two answer selections. See figure 1 for an example. The version shown to participants did not have the English text. The order of questions was randomized, and the answer selections were also presented in random order.

経営者側は従業員に昇給させた。彼らは他の選択肢がなかったからだ。
Management gave the workers a raise because they had no other choice.

選択肢がなかったのは誰ですか？
Who had no other choice?

☐ 従業員 workers
☐ 経営者側 management

Figure 1: Example stimuli for task 1 (pronoun resolution)

Table 2: Example of overt and null pronoun versions used

Overt Pronoun (Original)	Null Pronoun
教授は学生に低い成績をつけた。 <u>彼は</u> 厳格だったからだ。 厳格だったのは誰ですか? The professor gave the student a bad grade because <u>he</u> was strict. Who was strict? Answer: the professor	教授は学生に低い成績をつけた。 <u>Ø</u> 厳格だったからだ。 厳格だったのは誰ですか? The professor gave the student a bad grade because <u>Ø</u> was strict. Who was strict? Answer: the professor
教授は学生に低い成績をつけた。 <u>彼が</u> 勉強しなかった からだ。勉強しなかったのは誰ですか? The professor gave the student a bad grade because <u>he</u> did not study. Who did not study? Answer: the student	教授は学生に低い成績をつけた。 <u>Ø</u> 勉強しなかった からだ。勉強しなかったのは誰ですか? The professor gave the student a bad grade because <u>Ø</u> did not study. Who did not study? Answer: the student

In task 2, participants were shown one item at a time. Each item consisted of the two sentences from a WSC problem and a prompt asking the participant to rate the sentences on a five-point likert scale for naturalness, where 1 is considered very unnatural and 5 very natural. The order of questions was randomized. See figure 2 for an example. The English text was not shown to participants.

経営者側は従業員に昇給させた。彼らは他の選択肢がなかったからだ。
Management gave the workers a raise because they had no other choice.

上記の文章が日本語としてどの程度自然か評価してください。
Please rate how natural these sentences are in Japanese.

- ☐ 1 - とても不自然 Very unnatural
☐ 2 - 不自然 Unnatural
☐ 3 - 自然でも不自然でもない Neither natural nor unnatural
☐ 4 - 自然 Natural
☐ 5 - とても自然 Very natural

Figure 2: Example stimuli for task 2 (naturalness judgement)

3.1.5 Results. For each question on the pronoun resolution task, participants' answers were coded as 1 if the pronoun is resolved correctly, and 0 otherwise. Averaging across the participants, we found that the participants achieved high accuracy level on both conditions with overt pronouns ($M_{human_overt} = 0.97$, $SD = 0.05$) and conditions with null pronouns ($M_{human_null} = 0.96$, $SD = 0.05$). The difference between the two conditions was not statistically significant on a paired t-test ($t(32) = 1$, $p = 0.28$, two-tailed). These results indicate that variation in the surface form – presence vs. absence of a pronoun – do not affect interpretation.

When analyzing naturalness judgements, we averaged participants' ratings for each question. Sentences with overt pronouns received higher naturalness rating ($M_{human_overt} = 3.28$, $SD = 0.53$) compared to the sentences with null pronouns ($M_{human_null} = 3.08$, $SD = 0.56$). The differences between the two conditions were statistically significant on a paired t-test ($t(32) = 4.27$, $p < 0.01$, two-tailed).

3.2 Study 2: ChatGPT performance on the Japanese Winograd Schema Challenge

The goals for experiment 2 were to test ChatGPT's accuracy on the Japanese WSC problems and get naturalness ratings to compare to the human data. While the newer model GPT-4 is also available, we performed this study on GPT-3.5.

3.2.1 Procedure. The design and materials are identical to study 1, except for the order in which items were presented to ChatGPT. We first presented all stimuli with the null pronoun condition (pronoun resolution and naturalness rating tasks), and then all stimuli in the overt pronoun condition (pronoun resolution and naturalness rating). Since the null pronoun condition is novel, ChatGPT's responses on this set are unlikely to be influenced by pre-training materials, and thus can more accurately reflect its true ability.

Prior to conducting the study, we experimented with zero-shot vs few-shot prompts using WSC questions that are not part of the test, but did not find any differences between the two formats. In what follows, we adopted the zero-shot format. In the task 1 sessions, ChatGPT was given the following instructions in Japanese: "Please answer the following questions with A or B." Then it was shown one problem at a time. Each problem consisted of two sentences, one question prompt and two answer selections labeled A and B. The questions were presented in random order and the order of the answers was also randomized.

In the task 2 sessions, ChatGPT was shown the same five-point likert scale prompt eliciting naturalness as the human participants. It was shown one item at a time in random order. While it was not asked to give an explanation, it provided a short passage explaining its reasoning for each answer. The reasonings provided are not considered in this study. The experiment was run twice and the results of the two runs were averaged.

3.2.2 Results. ChatGPT's accuracy on overt pronoun questions ($M_{ChatGPT_overt} = 0.80$) was slightly lower than on questions with null pronouns ($M_{ChatGPT_null} = 0.83$). In terms of naturalness judgement, ChatGPT rated sentences in the overt pronoun condition lower on the naturalness scale ($M_{ChatGPT_overt} = 3.48$) compared to the sentences with the null pronoun condition ($M_{ChatGPT_null} = 3.58$). We next used these means to compare human performance to ChatGPT's performance.

3.3 Comparing human and ChatGPT performance

The comparison of ChatGPT's performance on the WSC shows that on the pronoun resolution tasks, humans achieve better accuracy ($M_{human_overt} = 0.97$; $M_{human_null} = 0.96$) than ChatGPT on the same task ($M_{ChatGPT_overt} = 0.8$; $M_{ChatGPT_null} = 0.83$). The differences are statistically significant on a one sample t-test for

both the overt pronoun condition ($t(32) = 19.2$, $p < 0.01$, two-tailed) and the null pronoun condition ($t(32) = 15.6$, $p < 0.01$, two-tailed).

When evaluating results on the naturalness judgement task, we observe that humans judge Japanese WSC sentences as less natural ($M_{human_overt} = 3.28$; $M_{human_null} = 3.08$) compared to ChatGPT ($M_{ChatGPT_overt} = 3.46$; $M_{ChatGPT_null} = 3.58$). The differences are statistically significant on a one sample t-test in both the overt pronoun condition ($t(32) = -2.09$, $p = 0.04$, two-tailed) and in the null pronoun condition ($t(32) = -5.06$, $p < 0.01$, two-tailed).²

4 DISCUSSION

Our results demonstrate that there are differences in how ChatGPT performs on the Japanese WSC compared to human performance. Specifically, humans outperform ChatGPT (GPT-3.5) when using world knowledge and commonsense reasoning on pronoun resolution tasks. In our study, humans' overall accuracy was at the ceiling. Our results are comparable to the human baseline for English (92%), reported by [3]. Using this benchmark test to compare ChatGPT to humans gives us insight into how close ChatGPT's reasoning resembles that of the human participants. A system that performs better than chance can be said to demonstrate some commonsense thinking, but to be considered human level, it would have to match or exceed human performance [12]. We can conclude from this data that GPT-3.5's performance in Japanese lags behind human performance for now. This should inform our approach to handling LLM information processing and decision making.

Moreover, we also find differences in naturalness ratings of the Japanese WSC test sentences. Human participants' ratings of naturalness were surprisingly low. Such low ratings are reflective of the difficulties that the researchers face when translating a test developed in English to a typologically different language ([20], [4], [14], [2]). The low rating of naturalness by humans also calls into question whether this test is really appropriate for natural language processing benchmarks. In contrast to human judgement, ChatGPT's rating of naturalness were significantly higher. ChatGPT offered explanations for why it assigned the naturalness rating it did. While not discussed here, a study focusing on its explanations versus that of human participants could offer insights on the differences in reasoning between humans and AI. The fact that ChatGPT gives higher naturalness ratings suggests that it does not have a human-like intuition about what is natural Japanese.

Before we conclude, it is important to discuss the limitations of the study. First, to keep the survey to a reasonable time limit, the number of WSC items used in this study was only a small, curated fraction of the original dataset. Due to factors discussed in section 3.1.3, such as culturally specific knowledge and structural inconsistencies, human performance on the full dataset may be lower than on our selection. Second, the sample size was small at 33 responders. Third, a full analysis of the decision heuristics used by ChatGPT and humans in this study is not feasible due to space limitations, but is a necessary follow-up to this research. Finally, LLMs are quickly evolving. The results reported here were collected

using the GPT-3.5 model. It is possible that later versions of the model will achieve more human-like results.

5 CONCLUSION

The study contributes to the growing body of research on evaluating LLM performance on world knowledge and commonsense reasoning tasks and comparing it to human performance ([7], [11], [10], [8]). By focusing on Japanese, we expand the domain of evaluation of LLMs beyond English. The human data reported in this pilot study can serve as a preliminary baseline for future studies using the Japanese Winograd Schema Challenge.

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²As a reviewer correctly pointed out, there are limitations of the statistical analysis presented here. We chose one-sample t-test in our comparison of ChatGPT and human data because in our design ChatGPT does not provide sufficient variance, so we treat ChatGPT as a point-estimate. While there are different ways to approach the data, the results are rather strong and we are confident in our conclusions.

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A SUPPLEMENTARY MATERIALS

A.1 Examples of rejected items

See table 3.

A.2 List of stimuli used in experiment 1 and 2

See table 4.

A.3 Instructions used for task 1 and 2

第一部 Part 1

アンケートの第一部では、2つの日本語の文章と、その文章の内容に関する質問と2つの答えが表示されています。2つの答えの内正しいと思う方を選んでください。どちらの答えも妥当と思われる場合は、最も適切と思われる方を選んでください。

In Part 1 of the survey, you will be shown two Japanese sentences and a question with two answers about the content of the sentences. Please choose the answer you think is correct. If both options seem right, please pick the one you think is the most fitting.

Figure 3: Instructions given for task 1. The English text was not shown to participants.

第二部 Part 2

アンケートの第二部では、2つの日本語の文章が表示されます。それがどの程度自然かを5段階で評価してもらいます。1が非常に不自然で、5が非常に自然と言う評価になります。

In Part 2 of the survey, you will be shown two Japanese sentences and asked to rate how natural they sound on a scale of five, one being very unnatural and five being very natural.

Figure 4: Instructions given for task 2. The English text was not shown to participants.

Table 3: Examples of Rejected Items

Item	Reason for Rejection
ペイパルは不満を持つ客のために毎年数百万件の争議を解決しなくてはならない。彼らは取引保護策をとっているからだ。 Paypal has to settle millions of disputes every year by disgruntled users because they provide a transaction protection policy.	Comprised of more than just two simple clauses.
エイドリアン・ピーターソンはレジー・ブッシュよりタッチダウンの数が多い。彼はボールを前に運ぶのがあまり上手くないからだ。 Adrian Peterson has more touchdowns than Reggie Bush because he cannot run the ball as well.	Contains long, probably unfamiliar foreign names and requires cultural knowledge of the rules of American football.
ダン は ハリーの車を改造した。彼が丁寧に頼んだからだ。 Dan modified Harry’s car because he asked politely.	The referent of the target pronoun for resolution is “Harry”, who is neither the subject, object or indirect object of the first sentence.
父は息子にいつか騎士になれると説得した。彼は彼の夢を打ち砕きたくなかったからだ。 The father convinced his son that it is possible for him to one day become a knight, because he did not want to crush his dreams.	There is an ambiguous pronoun (he) and an ambiguous possessive article (his).
クーガーはウサギと友達になろうとした。それが皆に愛されていたからだ。 The cougar tried to befriend the rabbit because it was loved by everybody.	The first sentence is active voice and the second sentence is passive.

Table 4: List of stimuli used in experiment 1 and 2

Question	Answer
経営者側は従業員に昇給させた。彼らは他の選択肢がなかったからだ。選択肢がなかったのは誰ですか? Management gave the workers a raise because they had no other choice. Who had no other choice?	経営者側 management
経営者側は従業員に昇給させた。他の選択肢がなかったからだ。他の選択肢がなかったのは誰ですか? Management gave the workers a raise because (they) had no other choice. Who had no other choice?	経営者側 management
経営者側は従業員に昇給させた。彼らはストライキをすると脅したからだ。ストライキをすると脅したのは誰ですか? Management gave the workers a raise because they had threatened to strike. Who had threatened to strike?	従業員 the workers
経営者側は従業員に昇給させた。ストライキをすると脅したからだ。ストライキをすると脅したのは誰ですか? Management gave the workers a raise because (they) had threatened to strike. Who had threatened to strike?	従業員 the workers
ジョンはビルに聞いた。彼は頭がこんがらがってしまったからだ。頭がこんがらがってしまったのは誰ですか? John asked Bill because he was confused. Who was confused?	ジョン John
ジョンはビルに聞いた。頭がこんがらがってしまったからだ。頭がこんがらがってしまったのは誰ですか? John asked Bill because (he) was confused. Who was confused?	ジョン John
ジョンはビルに聞いた。彼が答えを知っていたからだ。答えを知っていたのは誰ですか? John asked Bill because he knew the answer. Who knew the answer?	ビル Bill
ジョンはビルに聞いた。答えを知っていたからだ。答えを知っていたのは誰ですか? John asked Bill because (he) knew the answer. Who knew the answer?	ビル Bill
民衆は政府に抗議した。彼らは飢えていたからだ。飢えていたのは誰ですか? The people protested the government because they were hungry. Who was hungry?	民衆 the people
民衆は政府に抗議した。飢えていたからだ。飢えていたのは誰ですか? The people protested the government because (they) were hungry. Who was hungry?	民衆 the people
民衆は政府に抗議した。彼らは増税しようとしていたからだ。増税しようとしていたのは誰ですか? The people protested the government because they were raising taxes. Who was raising taxes?	政府 the government
民衆は政府に抗議した。増税しようとしていたからだ。増税しようとしていたのは誰ですか? The people protested the government because (they) were raising taxes. Who was raising taxes?	政府 the government
男は少年にキャンディをあげると言った。しかし彼はキャンディをもっていなかった。キャンディをもっていなかったのは誰ですか? The man offered the boy candy, but he had no candy. Who had no candy?	男 the man
男は少年にキャンディをあげると言った。しかしキャンディをもっていなかった。キャンディをもっていなかったのは誰ですか? The man offered the boy candy, but (he) had no candy. Who had no candy?	男 the man
男は少年にキャンディをあげると言った。しかし彼はキャンディをもらわなかった。キャンディをもらわなかったのは誰ですか? The man offered the boy candy, but he got no candy. Who got no candy?	少年 the boy
男は少年にキャンディをあげると言った。しかしキャンディをもらわなかった。キャンディをもらわなかったのは誰ですか? The man offered the boy candy, but (he) got no candy. Who got no candy?	少年 the boy
教授は学生に低い成績をつけた。彼は厳格だったからだ。厳格だったのは誰ですか? The professor gave the student a bad grade because he was strict. Who was strict?	教授 the professor
教授は学生に低い成績をつけた。厳格だったからだ。厳格だったのは誰ですか? The professor gave the student a bad grade because (he) was strict. Who was strict?	教授 the professor
教授は学生に低い成績をつけた。彼が勉強しなかったからだ。勉強しなかったのは誰ですか? The professor gave the student a bad grade because he did not study. Who did not study?	学生 the student
教授は学生に低い成績をつけた。勉強しなかったからだ。勉強しなかったのは誰ですか? The professor gave the student a bad grade because (he) did not study. Who did not study?	学生 the student

Question	Answer
ボールは窓ガラスを割った。それが堅かったからだ。堅かったのは何ですか？	ボール
The ball broke the window because it was hard. What was hard?	the ball
ボールは窓ガラスを割った。堅かったからだ。堅かったのは何ですか？	ボール
The ball broke the window because (it) was hard. What was hard?	the ball
ボールは窓ガラスを割った。それが壊れやすかったからだ。壊れやすかったのは何ですか？	窓ガラス
The ball broke the window because it was fragile. What was fragile?	the window
ボールは窓ガラスを割った。壊れやすかったからだ。壊れやすかったのは何ですか？	窓ガラス
The ball broke the window because (it) was fragile. What was fragile?	the window
ネコはネズミを捕まえた。それは賢かったから。賢かったのは何ですか？	ネコ
The cat caught the mouse because it was clever. What was clever?	the cat
ネコはネズミを捕まえた。賢かったから。賢かったのは何ですか？	ネコ
The cat caught the mouse because (it) was clever. What was clever?	the cat
ネコはネズミを捕まえた。それは油断していたからだ。油断していたのは何ですか？	ネズミ
The cat caught the mouse because it was careless. What was careless?	the mouse
ネコはネズミを捕まえた。油断していたからだ。油断していたのは何ですか？	ネズミ
The cat caught the mouse because (it) was careless. What was careless?	the mouse
ジョンは病院にルークを見舞った。彼は近くに住んでいたからだ。近くに住んでいたのは誰ですか？	ジョン
John visited Luke in the hospital because he lived close by. Who lived close by?	John
ジョンは病院にルークを見舞った。近くに住んでいたからだ。近くに住んでいたのは誰ですか？	ジョン
John visited Luke in the hospital because (he) lived close by. Who lived close by?	John
ジョンは病院にルークを見舞った。彼は病気だったのだ。病気だったのは誰ですか？	ルーク
John visited Luke in the hospital because he was sick. Who was sick?	Luke
ジョンは病院にルークを見舞った。病気だったのだ。病気だったのは誰ですか？	ルーク
John visited Luke in the hospital because (he) was sick. Who was sick?	Luke
ショーンはジョーを叩いた。しかし彼は謝らなかった。謝らなかったのは誰ですか？	ショーン
Shawn hit Joe but he did not apologize. Who did not apologize?	Shawn
ショーンはジョーを叩いた。しかし謝らなかった。謝らなかったのは誰ですか？	ショーン
Shawn hit Joe but (he) did not apologize. Who did not apologize?	Shawn
ショーンはジョーを叩いた。しかし彼は叫び声を上げなかった。叫び声を上げなかったのは誰ですか？	ジョー
Shawn hit Joe but he did not scream. Who did not scream?	ジョー
ショーンはジョーを叩いた。しかし叫び声を上げなかった。叫び声を上げなかったのは誰ですか？	ジョー
Shawn hit Joe but (he) did not scream. Who did not scream?	Joe
ジョンはダイブを首にした。彼はあまりに多くの従業員を抱えていた。あまりに多くの従業員を抱えていたのは誰ですか？	ジョン
John fired Dave because he had too many employees. Who had too many employees?	John
ジョンはダイブを首にした。あまりに多くの従業員を抱えていた。あまりに多くの従業員を抱えていたのは誰ですか？	ジョン
John fired Dave because (he) had too many employees. Who had too many employees?	John
ジョンはダイブを首にした。彼は欠勤が多すぎた。欠勤が多すぎたのは誰ですか？	ダイブ
John fired Dave because he had too many absences. Who had too many absences?	Dave
ジョンはダイブを首にした。欠勤が多すぎた。欠勤が多すぎたのは誰ですか？	ダイブ
John fired Dave because (he) had too many absences. Who had too many absences?	Dave